

JEE Mains 2019 Chapter wise Question Bank

Three Dimensional Geometry – Questions

Q1

The equation of the line passing through $(-4, 3, 1)$, parallel to the plane $x + 2y - z - 5 = 0$ and intersecting

the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1}$ is:

(1) $\frac{x-4}{2} = \frac{y+3}{1} = \frac{z+1}{4}$

(2) $\frac{x+4}{1} = \frac{y-3}{1} = \frac{z-1}{3}$

(3) $\frac{x+4}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$

(4) $\frac{x+4}{-1} = \frac{y-3}{1} = \frac{z-1}{1}$

9 Jan Morning

Q2

The plane through the intersection of the planes $x + y + z = 1$ and $2x + 3y - z + 4 = 0$ and parallel to y -axis also passes through the point:

(1) $(-3, 0, -1)$ (2) $(-3, 1, 1)$

(3) $(3, 3, -1)$ (4) $(3, 2, 1)$

9 Jan Morning

Q3

The equation of the plane containing the straight line

$\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing

the straight lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$ is:

(1) $x - 2y + z = 0$

(2) $3x + 2y - 3z = 0$

(3) $x + 2y - 2z = 0$

(4) $5x + 2y - 4z = 0$

9 Jan Evening

Q4

The plane passing through the point $(4, -1, 2)$ and parallel

to the lines $\frac{x+2}{3} = \frac{y-2}{-1} = \frac{z+1}{2}$ and $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{3}$

also passes through the point:

(1) $(1, 1, -1)$

(2) $(1, 1, 1)$

(3) $(-1, -1, -1)$

(4) $(-1, -1, 1)$

10 Jan Morning

Q5

Let A be a point on the line

$\vec{r} = (1 - 3\mu)\hat{i} + (\mu - 1)\hat{j} + (2 + 5\mu)\hat{k}$ and

B(3, 2, 6) be a point in the space. Then the value of μ for which the vector $\overrightarrow{AB}$ is parallel to the plane $x - 4y + 3z = 1$ is:

(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) $\frac{1}{2}$

(4) $-\frac{1}{4}$

Three Dimensional Geometry

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Q6

The plane which bisects the line segment joining the points $(-3, -3, 4)$ and $(3, 7, 6)$ at right angles, passes through which one of the following points?

- (1) $(-2, 3, 5)$ (2) $(4, -1, 7)$
(3) $(2, 1, 3)$ (4) $(4, 1, -2)$

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Q7

On which of the following lines lies the point of inter-section of the line, $\frac{x-4}{2} = \frac{y-5}{2} = \frac{z-3}{1}$ and the plane, $x + y + z = 2$?

- (1) $\frac{x+3}{3} = \frac{4-y}{3} = \frac{z+1}{-2}$
(2) $\frac{x-4}{1} = \frac{y-5}{1} = \frac{z-5}{-1}$
(3) $\frac{x-1}{1} = \frac{y-3}{2} = \frac{z+4}{-5}$
(4) $\frac{x-2}{2} = \frac{y-3}{2} = \frac{z+3}{3}$

10 Jan Evening

Q8

The plane containing the line $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z-1}{3}$ and also containing its projection on the plane $2x + 3y - z = 5$, contains which one of the following points?

- (1) $(2, 2, 0)$ (2) $(-2, 2, 2)$
(3) $(0, -2, 2)$ (4) $(2, 0, -2)$

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Q9

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The direction ratios of normal to the plane through the points $(0, -1, 0)$ and $(0, 0, 1)$ and making an angle $\frac{\pi}{4}$ with the plane $y - z + 5 = 0$ are :

- (1) $2, -1, 1$ (2) $2, \sqrt{2}, -\sqrt{2}$
(3) $\sqrt{2}, 1, -1$ (4) $2\sqrt{3}, 1, -1$

11 Jan Morning

Q10

Two lines $\frac{x-3}{1} = \frac{y+1}{3} = \frac{z-6}{-1}$ and $\frac{x+5}{7} = \frac{y-2}{-6} = \frac{z-3}{4}$ intersect at the point R. The reflection of R in the xy - plane has coordinates :

- (1) $(2, -4, -7)$ (2) $(2, 4, 7)$
(3) $(2, -4, 7)$ (4) $(-2, 4, 7)$

11 Jan Evening

Q11

If the point $(2, \alpha, \beta)$ lies on the plane which passes through the points $(3, 4, 2)$ and $(7, 0, 6)$ and is perpendicular to the plane $2x - 5y = 15$, then $2\alpha - 3\beta$ is equal to :

- (1) 12 (2) 7
(3) 5 (4) 17

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Q12

The perpendicular distance from the origin to the plane containing the two lines, $\frac{x+2}{3} = \frac{y-2}{5} = \frac{z+5}{7}$ and

$\frac{x-1}{1} = \frac{y-4}{4} = \frac{z+4}{7}$, is :

- (1) $11\sqrt{6}$ (2) $\frac{11}{\sqrt{6}}$
(3) 11 (4) $6\sqrt{11}$

12 Jan Morning

Q13

Three Dimensional Geometry

If an angle between the line, $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z-3}{-2}$ and the

plane, $x - 2y - kz = 3$ is $\cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$, then a value of k is

- (1) $\sqrt{\frac{5}{3}}$ (2) $\sqrt{\frac{3}{5}}$ (3) $-\frac{3}{5}$ (4) $-\frac{5}{3}$

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Q14

Let S be the set of all real values of λ such that a plane passing through the points $(-\lambda^2, 1, 1)$, $(1, -\lambda^2, 1)$ and $(1, 1, -\lambda^2)$ also passes through the point $(-1, -1, 1)$. Then S is equal to :

- (1) $\{\sqrt{3}\}$ (2) $\{\sqrt{3}, -\sqrt{3}\}$
(3) $\{1, -1\}$ (4) $\{3, -3\}$

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Q15

The equation of a plane containing the line of intersection of the planes $2x - y - 4 = 0$ and $y + 2z - 4 = 0$ and passing through the point $(1, 1, 0)$ is :

- (1) $x - 3y - 2z = -2$ (2) $2x - z = 2$
(3) $x - y - z = 0$ (4) $x + 3y + z = 4$

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Q16

The length of the perpendicular from the point $(2, -1, 4)$ on the straight line, $\frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1}$ is :

- (1) greater than 3 but less than 4
(2) less than 2
(3) greater than 2 but less than 3
(4) greater than 4

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Q17

The vector equation of the plane through the line of intersection of the planes $x + y + z = 1$ and $2x + 3y + 4z = 5$ which is perpendicular to the plane $x - y + z = 0$ is :

- (1) $\vec{r} \times (\hat{i} - \hat{k}) + 2 = 0$ (2) $\vec{r} \cdot (\hat{i} - \hat{k}) - 2 = 0$
(3) $\vec{r} \times (\hat{i} + \hat{k}) + 2 = 0$ (4) $\vec{r} \cdot (\hat{i} - \hat{k}) + 2 = 0$

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Q18

If a point R(4, y, z) lies on the line segment joining the points P(2, -3, 4) and Q(8, 0, 10), then distance of R from the origin is :

- (1) $2\sqrt{14}$ (2) $2\sqrt{21}$ (3) 6 (4) $\sqrt{53}$

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Q19

If the line, $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ meets the plane,

$x + 2y + 3z = 15$ at a point P, then the distance of P from the origin is:

- (1) $\sqrt{5}/2$ (2) $2\sqrt{5}$ (3) 9/2 (4) 7/2

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Q20

A plane passing through the points $(0, -1, 0)$ and $(0, 0, 1)$ and making an angle $\frac{\pi}{4}$ with the plane $y - z + 5 = 0$, also passes through the point:

- (1) $(-\sqrt{2}, 1, -4)$ (2) $(\sqrt{2}, -1, 4)$
(3) $(-\sqrt{2}, -1, -4)$ (4) $(\sqrt{2}, 1, 4)$

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Q21

The vertices B and C of a ΔABC lie on the line, $\frac{x+2}{3} = \frac{y-1}{0} = \frac{z}{4}$ such that $BC = 5$ units. Then the area (in sq. units) of this triangle, given that the point A(1, -1, 2), is:

- (1) $5\sqrt{17}$ (2) $2\sqrt{34}$ (3) 6 (4) $\sqrt{34}$

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Q22

Three Dimensional Geometry

Let P be the plane, which contains the line of intersection of the planes, $x + y + z - 6 = 0$ and $2x + 3y + z + 5 = 0$ and it is perpendicular to the xy -plane. Then the distance of the point $(0, 0, 256)$ from P is equal to:

- (1) $17/\sqrt{5}$ (2) $63\sqrt{5}$
(3) $205\sqrt{5}$ (4) $11/\sqrt{5}$

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Q23

If Q $(0, -1, -3)$ is the image of the point P in the plane $3x - y + 4z = 2$ and R is the point $(3, -1, -2)$, then the area (in sq. units) of ΔPQR is :

- (1) $2\sqrt{13}$ (2) $\frac{\sqrt{91}}{4}$ (3) $\frac{\sqrt{91}}{2}$ (4) $\frac{\sqrt{65}}{2}$

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Q24

Let A $(3, 0, -1)$, B $(2, 10, 6)$ and C $(1, 2, 1)$ be the vertices of a triangle and M be the midpoint of AC. If G divides BM in the ratio, $2 : 1$, then $\cos (\angle GOA)$ (O being the origin) is equal to :

- (1) $\frac{1}{2\sqrt{15}}$ (2) $\frac{1}{\sqrt{15}}$
(3) $\frac{1}{6\sqrt{10}}$ (4) $\frac{1}{\sqrt{30}}$

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Q25

If the length of the perpendicular from the point $(\beta, 0, \beta)$ ($\beta \neq 0$) to the line, $\frac{x}{1} = \frac{y-1}{0} = \frac{z+1}{-1}$ is $\sqrt{\frac{3}{2}}$, then β is equal to:

- (1) 1 (2) 2 (3) -1 (4) -2

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Q26

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A perpendicular is drawn from a point on the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ to the plane $x + y + z = 3$ such that the foot of the perpendicular Q also lies on the plane $x - y + z = 3$. Then the co-ordinates of Q are :

- (1) $(1, 0, 2)$ (2) $(2, 0, 1)$
(3) $(-1, 0, 4)$ (4) $(4, 0, -1)$

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Q27

If the plane $2x - y + 2z + 3 = 0$ has the distances $\frac{1}{2}$ and $\frac{2}{3}$ units from the planes $4x - 2y + 4z + \lambda = 0$ and $2x - y + 2z + \mu = 0$, respectively, then the maximum value of $\lambda + \mu$ is equal to :

- (1) 9 (2) 15 (3) 5 (4) 13

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Q28

If the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the plane $2x + 3y - z + 13 = 0$ at a point P and the plane $3x + y + 4z = 16$ at a point Q, then PQ is equal to :

- (1) 14 (2) $\sqrt{14}$ (3) $2\sqrt{7}$ (4) $2\sqrt{14}$

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Q29

A plane which bisects the angle between the two given planes $2x - y + 2z - 4 = 0$ and $x + 2y + 2z - 2 = 0$, passes through the point :

- (1) $(1, -4, 1)$ (2) $(1, 4, -1)$
(3) $(2, 4, 1)$ (4) $(2, -4, 1)$

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Q30

The length of the perpendicular drawn from the point $(2, 1, 4)$ to the plane containing the lines

$$\vec{r} = (\hat{i} + \hat{j}) + \lambda(\hat{i} + 2\hat{j} - \hat{k}) \text{ and } \vec{r} = (\hat{i} + \hat{j}) + \mu(-\hat{i} + \hat{j} - 2\hat{k})$$

is :

- (1) 3 (2) $\frac{1}{3}$ (3) $\sqrt{3}$ (4) $\frac{1}{\sqrt{3}}$

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Three Dimensional Geometry - Solutions

Q1

(3) Let any point on the intersecting line

$$\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1} = \lambda \text{ (say)}$$

is $(-3\lambda - 1, 2\lambda + 3, -\lambda + 2)$ Since, the above point lies on a line which passes through the point $(-4, 3, 1)$

Then, direction ratio of the required line

$$= \langle -3\lambda - 1 + 4, 2\lambda + 3 - 3, -\lambda + 2 - 1 \rangle$$

$$\text{or } \langle -3\lambda + 3, 2\lambda, -\lambda + 1 \rangle$$

Since, line is parallel to the plane

$$x + 2y - z - 5 = 0$$

Then, perpendicular vector to the line is $\hat{i} + 2\hat{j} - \hat{k}$

$$\text{Now } (-3\lambda + 3)(1) + (2\lambda)(2) + (-\lambda + 1)(-1) = 0$$

$$\Rightarrow \lambda = -1$$

Now direction ratio of the required line = $\langle 6, -2, 2 \rangle$ or $\langle 3, -1, 1 \rangle$

Hence required equation of the line is

$$\frac{(x+4)}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$$

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Q2

(4) Since, equation of plane through intersection of planes

$$x + y + z = 1 \text{ and } 2x + 3y - z + 4 = 0 \text{ is}$$

$$(2x + 3y - z + 4) + \lambda(x + y + z - 1) = 0$$

$$(2 + \lambda)x + (3 + \lambda)y + (-1 + \lambda)z + (4 - \lambda) = 0 \quad \dots(1)$$

But, the above plane is parallel to y -axis then

$$(2 + \lambda) \times 0 + (3 + \lambda) \times 1 + (-1 + \lambda) \times 0 = 0$$

$$\Rightarrow \lambda = -3$$

Hence, the equation of required plane is

$$-x - 4z + 7 = 0$$

$$\Rightarrow x + 4z - 7 = 0$$

Therefore, $(3, 2, 1)$ the passes through the point.

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Q3

(1) Let the direction ratios of the plane containing lines

$$\frac{x}{3} = \frac{y}{4} = \frac{z}{2} \text{ and } \frac{x}{4} = \frac{y}{2} = \frac{z}{3} \text{ is } \langle a, b, c \rangle$$

$$\therefore 3a + 4b + 2c = 0$$

$$4a + 2b + 3c = 0$$

$$\therefore \frac{a}{12-4} = \frac{b}{8-9} = \frac{c}{6-16}$$

$$\frac{a}{8} = \frac{b}{-1} = \frac{c}{-10}$$

$$\therefore \text{Direction ratio of plane} = \langle -8, 1, 10 \rangle.$$

Let the direction ratio of required plane is $\langle l, m, n \rangle$

$$\text{Then } -8l + m + 10n = 0 \quad \dots(1)$$

$$\text{and } 2l + 3m + 4n = 0 \quad \dots(2)$$

From (1) and (2),

$$\frac{l}{-26} = \frac{m}{52} = \frac{n}{-26}$$

$$\therefore \text{D.R.s are } \langle 1, -2, 1 \rangle$$

$$\therefore \text{Equation of plane: } x - 2y + z = 0$$

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Q4

(2) Equation of required plane is

$$\begin{vmatrix} x-4 & y+1 & z-2 \\ 3 & -1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$(x-4)(-3-4) - (y+1)(9-2) + (z-2)(6+1) = 0$$

$$\Rightarrow -7(x-4) - 7(y+1) + 7(z-2) = 0$$

$$\Rightarrow x - 4 + y + 1 - z + 2 = 0$$

$$\Rightarrow x + y - z - 1 = 0$$

$$\therefore \text{point } (1, 1, 1) \text{ satisfies this equation}$$

$$\therefore \text{point } (1, 1, 1) \text{ lies on the plane}$$

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Three Dimensional Geometry

Q5

(1) $\therefore A$ be a point on given line.

$\therefore$ Position vector of A

$$\overrightarrow{OA} = \hat{r} = (1 - 3\mu)\hat{i} + (\mu - 1)\hat{j} + (2 + 5\mu)\hat{k}$$

$$\text{Position vector of } B = \overrightarrow{OB} = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

$$\therefore \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= (3\mu + 2)\hat{i} + (3 - \mu)\hat{j} + (4 - 5\mu)\hat{k}$$

$$\text{Equation of plane is: } x - 4y + 3z = 1$$

$\therefore \overrightarrow{AB}$ is parallel to this plane.

$$\therefore 1(3\mu + 2) - 4(3 - \mu) + 3(4 - 5\mu) = 0$$

$$\Rightarrow 3\mu + 2 - 12 + 4\mu + 12 - 15\mu = 0$$

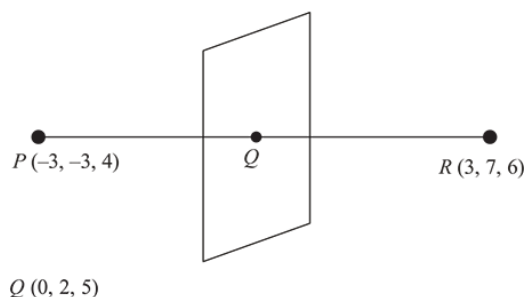
$$\Rightarrow 2 - 8\mu = 0$$

$$\Rightarrow \mu = \frac{1}{4}$$

Q6

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(4)



Since, direction ratios of normal to the plane is

$$\vec{n} = 6\hat{i} + 10\hat{j} + 2\hat{k}$$

Then, equation of the plane is

$$(x - 0)6 + (y - 2)10 + (z - 5)2 = 0$$

$$3x + 5y - 10 + z - 5 = 0$$

$$3x + 5y + z = 15 \quad \dots(1)$$

Since, plane (1) satisfies the point (4, 1, -2)

Hence, required point is (4, 1, -2)

Q7

(3) Let any point on the line $\frac{x-4}{2} = \frac{y-5}{2} = \frac{z-3}{1}$ be $A(2\lambda + 4, 2\lambda + 5, \lambda + 3)$ which lies on the plane $x + y + z = 2$

$$\Rightarrow 2\lambda + 4 + 2\lambda + 5 + \lambda + 3 = 2$$

$$\Rightarrow 5\lambda = -10 \Rightarrow \lambda = -2$$

Then, the point of intersection is (0, 1, 1)

$$\text{which lies on the line } \frac{x-1}{1} = \frac{y-3}{2} = \frac{z+4}{-5}$$

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Q8

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Three Dimensional Geometry

(4) Let normal to the required plane is $\hat{n}$

$\Rightarrow \hat{n}$ is perpendicular to both vector $2\hat{i} - \hat{j} + 3\hat{k}$ and $2\hat{i} + 3\hat{j} - 3\hat{k}$.

$$\Rightarrow \hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 3 \\ 2 & 3 & -1 \end{vmatrix} = -8\hat{i} + 8\hat{j} + 8\hat{k}$$

$\Rightarrow$ Equation of the required plane is

$$\Rightarrow (x-3)(-8) + (y+2) \times 8 + (z-1) \times 8 = 0$$

$$\Rightarrow (x-3)(-1) + (y+2) \times 1 + (z-1) \times 1 = 0$$

$$\Rightarrow x - 3 - y - 2 - z + 1 = 0$$

$$\therefore x - y - z = 4 \text{ passes through } (2, 0, -2)$$

$$\therefore \text{plane contains } (2, 0, -2).$$

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Q9

(2, 3) Let the d.r's of the normal be $\langle a, b, c \rangle$

Equation of the plane is

$$a(x-0) + b(y+1) + c(z-0) = 0$$

$\therefore$ It passes through $(0, 0, 1)$

$$\therefore b + c = 0$$

$$\text{Also } \frac{0 \cdot a + b - c}{\sqrt{a^2 + b^2 + c^2} \cdot \sqrt{2}} = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow b - c = \sqrt{a^2 + b^2 + c^2}$$

$$\text{And } b + c = 0$$

$$\Rightarrow b = \pm \frac{1}{\sqrt{2}} a.$$

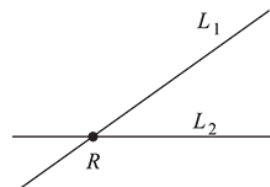
$$\therefore \text{The d.r's are } \sqrt{2}, 1, -1 \text{ or } 2, \sqrt{2}, -\sqrt{2}$$

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Q10

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(1) Let the coordinate of P with respect to line



$$\frac{x-3}{1} = \frac{y+1}{3} = \frac{z-6}{-1} = \lambda$$

$$\frac{x+5}{7} = \frac{y-2}{-6} = \frac{z-3}{4} = \mu$$

$$L_1 = (\lambda + 3, 3\lambda - 1, -\lambda + 6)$$

and coordinate of P w.r.t.

$$\text{line } L_2 = (7\mu - 5, -6\mu + 2, 4\mu + 3)$$

$$\therefore \lambda - 7\mu = -8, 3\lambda + 6\mu = 3, \lambda + 4\mu = 3$$

$$\text{From above equation : } \lambda = -1, \mu = 1$$

$$\therefore \text{Coordinate of point of intersection } R = (2, -4, 7).$$

$$\text{Image of } R \text{ w.r.t. } xy \text{ plane} = (2, -4, -7).$$

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Q11

(2) Let the normal to the required plane is n , then

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -4 & 4 \\ 2 & -5 & 0 \end{vmatrix} = 20\hat{i} + 8\hat{j} - 12\hat{j}$$

$\therefore$ Equation of the plane

$$(x-3) \times 20 + (y-4) \times 8 + (z-2) \times (-12) = 0$$

$$5x - 15 + 2y - 8 - 3z + 6 = 0$$

$$5x + 2y - 3z - 17 = 0 \quad \dots(1)$$

Since, equation of plane (1) passes through $(2, \alpha, \beta)$, then

$$10 + 2\alpha - 3\beta - 17 = 0 \Rightarrow 2\alpha - 3\beta = 7$$

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Q12

Three Dimensional Geometry

(2) $\therefore$ plane containing both lines.

$$\therefore \text{D.R. of plane} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & 7 \\ 1 & 4 & 7 \end{vmatrix} = 7\hat{i} - 14\hat{j} + 7\hat{k}$$

Now, equation of plane is,

$$7(x-1) - 14(y-4) + 7(z+4) = 0$$

$$\Rightarrow x - 1 - 2y + 8 + z + 4 = 0$$

$$\Rightarrow x - 2y + z + 11 = 0$$

Hence, distance from (0, 0, 0) to the plane,

$$= \frac{11}{\sqrt{1+4+1}} = \frac{11}{\sqrt{6}}$$

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Q13

(1) Let angle between line and plane is θ , then

$$\sin \theta = \frac{\left| \frac{\mathbf{r}}{b} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|}{\left| \frac{\mathbf{r}}{b} \right| \cdot \left| \frac{\mathbf{n}}{|\mathbf{n}|} \right|} = \frac{|(2\hat{i} + \hat{j} - 2\hat{k}) \cdot (\hat{i} - 2\hat{j} - K\hat{k})|}{\sqrt{9} \cdot \sqrt{1+4+K^2}}$$

$$= \frac{|2-2-2K|}{3\sqrt{5+K^2}} = \frac{2|K|}{3\sqrt{5+K^2}}$$

$$\text{Since, } \cos \theta = \frac{2\sqrt{2}}{3} \Rightarrow \sin \theta = \frac{1}{3}$$

$$\text{Then, } \frac{2|K|}{3\sqrt{5+K^2}} = \frac{1}{3} \Rightarrow 4K^2 = 5 + K^2$$

$$3K^2 = 5 \Rightarrow K = \pm \sqrt{\frac{5}{3}}$$

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Q14

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(2) Let $A(-\lambda^2, 1, 1)$, $B(1, -\lambda^2, 1)$, $C(1, 1, -\lambda^2)$, $D(-1, -1, 1)$ lie on same plane, then

$$\begin{vmatrix} 1-\lambda^2 & 2 & 0 \\ 2 & 1-\lambda^2 & 0 \\ 2 & 2 & -\lambda^2-1 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda^2 + 1)((1 - \lambda^2)^2 - 4) = 0$$

$$\Rightarrow (3 - \lambda^2)(\lambda^2 + 1) = 0 \Rightarrow \lambda^2 = 3$$

$$\lambda = \pm\sqrt{3}$$

$$\text{Hence, } S = \{-\sqrt{3}, \sqrt{3}\}$$

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Q15

(3) Let the equation of required plane be;

$$(2x - y - 4) + \lambda(y + 2z - 4) = 0$$

$\therefore$ This plane passes through the point (1, 1, 0) then $(2 - 1 - 4) + \lambda(1 + 0 - 4) = 0$

$$\Rightarrow \lambda = -1$$

Then, equation of required plane is,

$$(2x - y - 4) - (y + 2z - 4) = 0$$

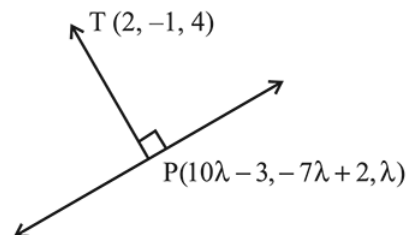
$$\Rightarrow 2x - 2y - 2z = 0 \Rightarrow x - y - z = 0$$

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Q16

Three Dimensional Geometry

- (1) Let P be the foot of perpendicular from point $T(2, -1, 4)$ on the given line. So P can be assumed as $P(10\lambda - 3, -7\lambda + 2, \lambda)$



DR's of TP is proportional to $10\lambda - 5, -7\lambda + 3, \lambda - 4$
 $\therefore TP$ and given line are perpendicular, so

$$10(10\lambda - 5) - 7(-7\lambda + 3) + 1(\lambda - 4) = 0$$

$$\Rightarrow \lambda = \frac{1}{2}$$

$$\Rightarrow TP = \sqrt{(10\lambda - 5)^2 + (-7\lambda + 3)^2 + (\lambda - 4)^2}$$

$$= \sqrt{0 + \frac{1}{4} + \frac{49}{4}} = \sqrt{12.5} = 3.54$$

Hence, the length of perpendicular is greater than 3 but less than 4.

8 April Morning

Q17

- (4) Equation of the plane passing through the line of intersection of $x + y + z = 1$ and $2x + 3y + 4z = 5$ is
 $(2x + 3y + 4z - 5) + \lambda(x + y + z - 1) = 0$
 $\Rightarrow (2 + \lambda)x + (3 + \lambda)y + (4 + \lambda)z + (-5 - \lambda) = 0 \dots (i)$
 $\therefore$ plane (i) is perpendicular to the plane $x - y + z = 0$
 $\therefore (2 + \lambda)(1) + (3 + \lambda)(-1) + (4 + \lambda)(1) = 0$
 $2 + \lambda - 3 - \lambda + 4 + \lambda = 0 \Rightarrow \lambda = -3$
Hence, equation of required plane is
 $-x + z - 2 = 0$ or $x - z + 2 = 0$
 $\Rightarrow \vec{r} \cdot (\hat{i} - \hat{k}) + 2 = 0$

8 April Evening

Q18

- (1) Here, P, Q, R are collinear

$$\therefore \vec{PR} = \lambda \vec{PQ}$$

$$2\hat{i} + (y+3)\hat{j} + (z-4)\hat{k} = \lambda[6\hat{i} + 3\hat{j} + 6\hat{k}]$$

$$\Rightarrow 6\lambda = 2, y + 3 = 3\lambda, z - 4 = 6\lambda$$

$$\Rightarrow \lambda = \frac{1}{3}, y = -2, z = 6$$

$$\therefore \text{Point } R(4, -2, 6)$$

$$\text{Now, } OR = \sqrt{(4)^2 + (-2)^2 + (6)^2} = \sqrt{56} = 2\sqrt{14}$$

JEE Mains 2019 Chapter wise Question Bank

8 April Evening

Q19

- (3) Let point on line be $P(2k + 1, 3k - 1, 4k + 2)$
Since, point P lies on the plane $x + 2y + 3z = 15$
 $\therefore 2k + 1 + 6k - 2 + 12k + 6 = 15$

$$\Rightarrow k = \frac{1}{2}$$

$$\therefore P \equiv \left(2, \frac{1}{2}, 4\right)$$

Then the distance of the point P from the origin is

$$OP = \sqrt{4 + \frac{1}{4} + 16} = \frac{9}{2}$$

9 April Morning

Q20

- (4) Let the required plane passing through the points

$$(0, -1, 0) \text{ and } (0, 0, 1) \text{ be } \frac{x}{\lambda} + \frac{y}{-1} + \frac{z}{1} = 1 \text{ and the given}$$

$$\text{plane is } y - z + 5 = 0$$

$$\therefore \cos \frac{\pi}{4} = \frac{-1-1}{\sqrt{\left(\frac{1}{\lambda^2} + 1 + 1\right)} \sqrt{2}}$$

$$\Rightarrow \lambda^2 = \frac{1}{2} \Rightarrow \frac{1}{\lambda} = \pm \sqrt{2}$$

Then, the equation of plane is

$$\pm \sqrt{2}x - y + z = 1$$

Then the point $(\sqrt{2}, 1, 4)$ satisfies the equation of plane

$$-\sqrt{2}x - y + z = 1$$

9 April Morning

Q21

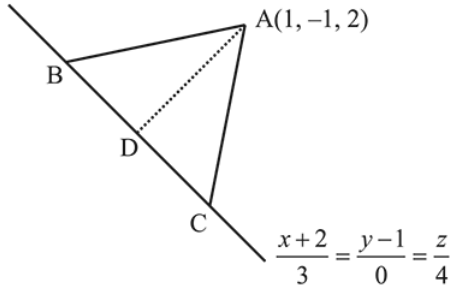
Three Dimensional Geometry

- (4) Let a point D on BC = $(3\lambda - 2, 1, 4\lambda)$

$$\overrightarrow{AD} = (3\lambda - 3)\hat{i} + 2\hat{j} + (4\lambda - 2)\hat{k}$$

$$\therefore \overrightarrow{AD} \perp \overrightarrow{BC}, \therefore \overrightarrow{AD} \cdot \overrightarrow{BC} = 0$$

$$\Rightarrow (3\lambda - 3) + 3 + 2(0) + (4\lambda - 2)4 = 0 \Rightarrow \lambda = \frac{17}{25}$$



$$\text{Hence, } D = \left(\frac{1}{25}, 1, \frac{68}{25} \right)$$

$$|\overrightarrow{AD}| = \sqrt{\left(\frac{1}{25} - 1 \right)^2 + (2)^2 + \left(\frac{68}{25} - 2 \right)^2}$$

$$= \sqrt{\frac{(24)^2 + 4(25)^2 + (18)^2}{25}} = \sqrt{\frac{3400}{25}} = \frac{2\sqrt{34}}{5}$$

$$\text{Area of triangle} = \frac{1}{2} \times |\overrightarrow{BC}| \times |\overrightarrow{AD}|$$

$$= \frac{1}{2} \times 5 \times \frac{2\sqrt{34}}{5} = \sqrt{34} \quad [\because BC = 5]$$

9 April Evening

Q22

- (4) Let the plane be

$$P \equiv (2x + 3y + z + 5) + \lambda(x + y + z - 6) = 0$$

$\therefore$ above plane is perpendicular to xy plane.

$$\therefore ((2+\lambda)\hat{i} + (3+\lambda)\hat{j} + (1+\lambda)\hat{k}) \cdot \hat{k} = 0 \Rightarrow \lambda = -1$$

Hence, the equation of the plane is,

$$P \equiv x + 2y + 11 = 0$$

Distance of the plane P from $(0, 0, 256)$

$$\left| \frac{0+0+11}{\sqrt{5}} \right| = \frac{11}{\sqrt{5}}$$

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Q23

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- (3) Image of Q $(0, -1, -3)$ in plane is,

$$\frac{(x-0)}{3} = \frac{(y+1)}{-1} = \frac{z+3}{+4} = \frac{-2(1-12-2)}{9+1+16} = 1$$

$$\Rightarrow x = 3, y = -2, z = 1$$

$$\Rightarrow P(3, -2, 1), Q(0, -1, -3), R(3, -1, -2)$$

$\therefore$ Area of ΔPQR is

$$\frac{1}{2} |\vec{QP} \times \vec{QR}| = \frac{1}{2} \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 4 \\ 3 & 0 & 1 \end{vmatrix} \right\|$$

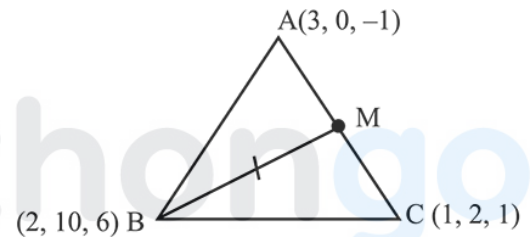
$$= \frac{1}{2} |\{\hat{i}(-1) - \hat{j}(3-12) + \hat{k}(3)\}|$$

$$= \frac{1}{2} \sqrt{(1+81+9)} = \frac{\sqrt{91}}{2}$$

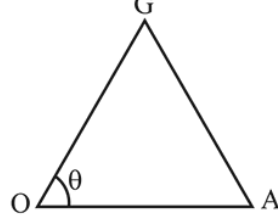
10 April Morning

Q24

- (2) G is the centroid of ΔABC .



$$\Rightarrow G \equiv \left(\frac{3+2+1}{3}, \frac{0+10+2}{3}, \frac{-1+6+1}{3} \right) \equiv (2, 4, 2)$$



$$OG = \sqrt{4+16+4}, OA = \sqrt{9+1}, AG = \sqrt{1+16+9}$$

$$\therefore \cos \theta = \frac{(OG)^2 + (OA)^2 - (AG)^2}{2(OG)(OA)} = \frac{24+10-26}{2\sqrt{24}\sqrt{10}}$$

$$= \frac{8}{2\sqrt{8 \times 3 \times 2 \times 5}} = \frac{4}{4\sqrt{15}} = \frac{1}{\sqrt{15}}$$

10 April Morning

Q25

Three Dimensional Geometry

- (3) Given, $\frac{x}{1} = \frac{y-1}{0} = \frac{z+1}{-1} = p$ (let) and point P (β , 0, β)

Any point on line A = (p , 1, $-p-1$)

Now, DR of AP $\equiv \langle p-\beta, 1-0, -p-1-\beta \rangle$

Which is perpendicular to line.

$$\therefore (p-\beta)1 + 0 \cdot 1 - 1(-p-1-\beta) = 0$$

$$\Rightarrow p - \beta + p + 1 + \beta = 0 \Rightarrow p = \frac{-1}{2}$$

$$\therefore \text{Point A} \left(\frac{-1}{2}, 1 - \frac{1}{2} \right)$$

$$\text{Given that distance AP} = \frac{\sqrt{3}}{2} \Rightarrow \text{AP}^2 = \frac{3}{2}$$

$$\Rightarrow \left(\beta + \frac{1}{2} \right)^2 + 1 + \left(\beta + \frac{1}{2} \right)^2 = \frac{3}{2} \text{ or } 2 \left(\beta + \frac{1}{2} \right)^2 = \frac{1}{2}$$

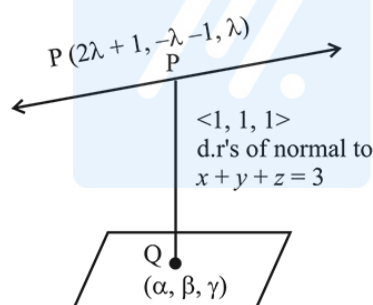
$$\Rightarrow \left(\beta + \frac{1}{2} \right)^2 = \frac{1}{4} \Rightarrow \beta = 0, -1, (\beta \neq 0)$$

$$\therefore \beta = -1$$

10 April Morning

Q26

(2)



Let co-ordinates of Q be (α , β , γ), then

$$\alpha + \beta + \gamma = 3 \quad \dots(i)$$

$$\alpha - \beta + \gamma = 3 \quad \dots(ii)$$

$$\Rightarrow \alpha + \gamma = 3 \text{ and } \beta = 0$$

Equating direction ratio's of PQ, we get

$$\frac{\alpha - 2\lambda - 1}{1} = \frac{\lambda + 1}{1} = \frac{\gamma - \lambda}{1}$$

$$\Rightarrow \alpha = 3\lambda + 2, \gamma = 2\lambda + 1$$

Substituting the values of α and γ in equation (i), we get

$$\Rightarrow 5\lambda + 3 = 3 \Rightarrow \lambda = 0$$

Hence, point is Q (2, 0, 1)

10 April Evening

Q27

JEE Mains 2019 Chapter wise Question Bank

- (4) Let, $P_1 : 2x - y + 2z + 3 = 0$

$$P_2 : 2x - y + 2z + \frac{\lambda}{2} = 0$$

$$P_3 : 2x - y + 2z + \mu = 0$$

Given, distance between P_1 and P_2 is $\frac{1}{3}$

$$\frac{1}{3} = \frac{\left| 3 - \frac{\lambda}{2} \right|}{\sqrt{9}} \Rightarrow \left| 3 - \frac{\lambda}{2} \right| = 1 \Rightarrow \lambda_{\max} = 8$$

And distance between P_1 and P_3 is $\frac{2}{3}$

$$\frac{2}{3} = \frac{|\mu - 3|}{\sqrt{9}} \Rightarrow \mu_{\max} = 5$$

$$\Rightarrow (\lambda + \mu)_{\max} = 13$$

10 April Evening

Q28

- (4) Let points P ($3\lambda + 2$, $2\lambda - 1$, $-\lambda + 1$) and

$$Q (3\mu + 2, 2\mu - 1, -\mu + 1)$$

$$\therefore P \text{ lies on } 2x + 3y - z + 13 = 0$$

$$\therefore 6\lambda + 4 + 6\lambda - 3 + \lambda - 1 + 13 = 0$$

$$\Rightarrow 13\lambda = -13 \Rightarrow \lambda = -1$$

Hence, P (-1, -3, 2)

Similarly, Q lies on $3x + y + 4z = 16$

$$\therefore 9\mu + 6 + 2\mu - 1 - 4\mu + 4 = 16$$

$$\Rightarrow 7\mu = 7 \Rightarrow \mu = 1$$

Hence, Q is (5, 1, 0)

$$\text{Now, } PQ = \sqrt{36 + 16 + 4} = \sqrt{56} = 2\sqrt{14}$$

12 April Morning

Q29

- (4) The equations of angle bisectors are,

$$\frac{x + 2y + 2z - 2}{3} = \pm \frac{2x - y + 2z - 4}{3}$$

$$\Rightarrow x - 3y - 2 = 0$$

$$\text{or } 3x + y + 4z - 6 = 0$$

(2, -4, 1) lies on the second plane.

12 April Evening

Q30

(3) The equation of plane containing two given lines is,

$$\begin{vmatrix} x-1 & y-1 & z \\ 1 & 2 & -1 \\ -1 & 1 & -2 \end{vmatrix} = 0$$

On expanding, we get $x - y - z = 0$

Now, the length of perpendicular from (2, 1, 4) to this plane

$$= \frac{|2 - 1 - 4|}{\sqrt{1^2 + 1^2 + 1^2}} = \sqrt{3}$$

12 April Evening



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